D07 · Warehouse Automation & AMR/Robotics

02 · Automated Storage & Retrieval Systems (ASRS)

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Automated Storage and Retrieval Systems (AS/RS) store and retrieve pallets, totes or cases with cranes, shuttles or robotic grids under software control, using building height to achieve very high storage density.

2 WHY IT MATTERS

AS/RS turn vertical space into capacity and deliver high accuracy and throughput with little labor. **They are also the most capital-intensive and least flexible option, and a single failure can stop the operation**. They make sense where land is expensive, volume is stable and high, and the business can commit to the design for many years.

3 HOW IT IS DONE

1. **Validate the density case.** Compare AS/RS with conventional racking, densification and additional leased space.
2. **Size for throughput and growth.** Peak movements per hour, not average, with headroom.
3. **Design a manual fallback.** An emergency process and space for when the system is down.
4. **Contract maintenance with SLAs.** Availability and response times written into the agreement.

IN PRACTICE

ASRS has the greatest ROI in warehouses where m² is the scarcest resource

In urban zones where m² cost is very high, ASRS can generate a 3–5 year ROI even with high capex.

Always design a manual operation alternative for when the ASRS fails

A warehouse that depends 100% on the ASRS and has no manual operation alternative has a total shutdown risk. The design must include minimum space for an emergency manual operation.

ASRS-WMS integration requires specialized middleware (WCS)

The WCS (Warehouse Control System) manages crane movement. Correct WMS-WCS integration ensures ASRS movements are coordinated with WMS picking orders.

The ASRS maintenance contract is as important as the system price

Preventive and corrective maintenance with availability and response time SLAs is the factor that determines the system's real useful life.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Pallet ASRS — pharmaceutical warehouse in urban zone, limited space — The warehouse needs to triple its capacity without expanding m².

Alternative	Capacity in 3,200 m²	Estimated capex
Additional conventional shelving	MEDIUM	2,100 pallets (vs. 1,800 current) · Only +17% capacity
Pallet ASRS at 28 meters height	OK	6,400 pallets (+256%) · No footprint expansion
vs. Additional 4,500 m² lease	HIGH	Equivalent · $7.8M in 5 years of urban zone lease

Result: Selection: Pallet ASRS at 28 meters. Capex: $4.2M. ROI vs. additional lease: $7.8M - $4.2M = $3.6M in 5 years. Payback: 3.2 years. 6,400-pallet capacity in the existing footprint was the decisive argument.

HOW IT IS MEASURED

ASRS Throughput (movements per hour)

ASRS ThroughputTotal storage and retrieval movements / ASRS operating hours

Benchmark: Pallet ASRS: 60–120 movements/hr · Mini-load: 200–500 movements/hr · AutoStore: 300–800 movements/hr Throughput below manufacturer specs may indicate software problems, mechanical failures, or a demand profile incompatible with the system design.

ASRS Availability % (% of time the ASRS is operational without failures)

ASRS Availability %(ASRS operating hours without failure / Total planned operating hours) × 100

Benchmark: >98% availability for ASRS in critical operations An ASRS with <95% availability in a pharmaceutical or high-speed e-commerce warehouse is a serious operational risk — the warehouse can be paralyzed if the ASRS fails without a manual alternative.

WHAT YOU WOULD USE

ASRS Providers

Dematic / SSI Schaefer — Module: Pallet ASRS + Mini-Load ASRSTwo of the largest warehouse automation providers globally.

AutoStore — Module: Grid-Based High-Density StorageVery high-density grid storage system. Especially suitable for pharmaceutical e-commerce and high-precision electronics.

4 SO WHAT

Build up only when the numbers and the volume are durable. Compare against alternatives, size for the peak, keep a manual fallback and contract availability, not just equipment.

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